

1. Hölder Inequality (赫爾德不等式)

(Otto Ludwig Hölder, 1859-1937, German mathematician)

For any two sequences of non-negative real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$,

and for any real numbers p and q such that $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$, (e.g. $p = \frac{3}{2}$, $q = 3$)

$$(a_1^p + a_2^p + \dots + a_n^p)^{\frac{1}{p}} (b_1^q + b_2^q + \dots + b_n^q)^{\frac{1}{q}} \geq (a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n)$$

The equality holds if and only if $\frac{a_1^p}{b_1^q} = \frac{a_2^p}{b_2^q} = \frac{a_3^p}{b_3^q} = \dots = \frac{a_n^p}{b_n^q}$.

Note That Cauchy's Inequality is a special case of Hölder Inequality when $p=q=2$.

Proof: For $i = 1, 2, \dots, n$, let $x_i = (a_i)^p$, $y_i = (b_i)^q$, and $A = \sum_{i=1}^n x_i$, $B = \sum_{i=1}^n y_i$,

and let $\alpha = \frac{1}{p}$, $\beta = \frac{1}{q}$, so $\alpha + \beta = 1$. $x_i = (a_i)^{\frac{1}{\alpha}}$, $y_i = (b_i)^{\frac{1}{\beta}}$.

Consider $f(x) = \ln x$, which is a strictly concave function in the interval $(0, \infty)$.

Using Jensen's Inequality, $\alpha \ln\left(\frac{x_i}{A}\right) + \beta \ln\left(\frac{y_i}{B}\right) \leq \ln\left\{\alpha\left(\frac{x_i}{A}\right) + \beta\left(\frac{y_i}{B}\right)\right\}$

$$\Rightarrow \left(\frac{x_i}{A}\right)^{\alpha} \left(\frac{y_i}{B}\right)^{\beta} \leq \left(\frac{\alpha x_i}{A} + \frac{\beta y_i}{B}\right) \quad \text{for } i = 1, 2, \dots, n.$$

$$\text{Summing up} \quad \sum_{i=1}^n \left(\frac{x_i}{A}\right)^{\alpha} \left(\frac{y_i}{B}\right)^{\beta} \leq \sum_{i=1}^n \left(\frac{\alpha x_i}{A} + \frac{\beta y_i}{B}\right) = \frac{\alpha}{A} \left(\sum_{i=1}^n x_i\right) + \frac{\beta}{B} \left(\sum_{i=1}^n y_i\right) = 1,$$

$$\text{which means that} \quad \sum_{i=1}^n x_i^{\alpha} y_i^{\beta} \leq A^{\alpha} B^{\beta} = \left(\sum_{i=1}^n x_i\right)^{\alpha} \left(\sum_{i=1}^n y_i\right)^{\beta}$$

The equality holds only when $\frac{x_i}{A} = \frac{y_i}{B}$ for $i = 1, 2, \dots, n$.

$$\text{That is} \quad \frac{x_1}{y_1} = \frac{x_2}{y_2} = \dots = \frac{x_n}{y_n}.$$

Remember that $x_i = (a_i)^{\frac{1}{\alpha}}$, $y_i = (b_i)^{\frac{1}{\beta}}$, $\alpha = \frac{1}{p}$, $\beta = \frac{1}{q}$, so $x_i = (a_i)^p$, $y_i = (b_i)^q$.

Hence $\sum_{i=1}^n a_i b_i \leq \left(\sum_{i=1}^n a_i^p\right)^{\frac{1}{p}} \left(\sum_{i=1}^n b_i^q\right)^{\frac{1}{q}}$, equality holds when $\frac{a_i^p}{b_i^q}$ has a common value for all 'i's.

Example 1: For any positive real numbers a, b and c , prove that

$$a\sqrt{\frac{a}{a+b}} + b\sqrt{\frac{b}{b+c}} + c\sqrt{\frac{c}{c+a}} \geq \frac{a+b+c}{\sqrt{2}}$$

Example 2: For any positive real numbers x, y and z , prove that

$$\left(\sum_{\text{cyclic}} (y+z) \sqrt{\frac{yz}{(z+x)(x+y)}}\right)^2 \left(\sum_{\text{cyclic}} y^2 z^2 (y+z)(z+x)(x+y)\right) \geq \left(\sum_{\text{cyclic}} yz(y+z)\right)^3$$

Extension:

For any three sequences of non-negative real numbers $a_1, a_2, a_3, \dots, a_n; b_1, b_2, b_3, \dots, b_n; c_1, c_2, c_3, \dots, c_n$ and for any real numbers p, q, r such that $p, q, r > 1$ and $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$,

$$(a_1^p + a_2^p + \dots + a_n^p)^{\frac{1}{p}} (b_1^q + b_2^q + \dots + b_n^q)^{\frac{1}{q}} (c_1^r + c_2^r + \dots + c_n^r)^{\frac{1}{r}} \geq (a_1 b_1 c_1 + a_2 b_2 c_2 + \dots + a_n b_n c_n)$$

Example 3: Let a, b, c be positive real numbers satisfying $a+b+c = 3$. What is the

minimum value of $\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$?

Example 4: (2001 IMO No. 2) Let a, b, c be positive real numbers. Prove that

$$\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ac}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1.$$

The general case is:

For non-negative real sequences $a_i, b_i, \dots, z_i$, and $\sigma_a + \sigma_b + \dots + \sigma_z = 1$,

$$(a_1 + a_2 + \dots + a_n)^{\sigma_a} (b_1 + b_2 + \dots + b_n)^{\sigma_b} \dots (z_1 + z_2 + \dots + z_n)^{\sigma_z} \geq a_1^{\sigma_a} b_1^{\sigma_b} \dots z_1^{\sigma_z} + a_2^{\sigma_a} b_2^{\sigma_b} \dots z_2^{\sigma_z} + \dots + a_n^{\sigma_a} b_n^{\sigma_b} \dots z_n^{\sigma_z}.$$

Example 5: Prove the AM-GM Inequality using the Hölder Inequality.

2. Schur's Inequality (舒爾不等式)

(Issai Schur, German Mathematician, 1875-1941)

If x, y, z are non-negative real numbers, and r a real number, then $\sum_{cyc} x^r (x-y)(x-z) \geq 0$.

Equality holds iff either $x=y=z$ or two of them are equal and the third zero.

Proof: Without loss of generality, assume that $x \geq y \geq z$, then

$$\begin{aligned} \sum_{cyc} x^r (x-y)(x-z) &= x^r(x-y)(x-z) + y^r(y-z)(y-x) + z^r(z-x)(z-y) \\ &= (x-y)[x^r(x-z) - y^r(y-z)] + z^r(x-z)(y-z) \\ &\geq (x-y)[y^r(x-z) - y^r(y-z)] + z^r(x-z)(y-z) \geq 0. \end{aligned}$$

Alternative Proof: Without loss of generality, assume that $x \geq y \geq z$, and let

$x = t_1 + t_2 + t_3, y = t_2 + t_3, z = t_3$, where t_1, t_2, t_3 are non-negative real numbers.

Then, $\sum_{cyc} x^r (x-y)(x-z) = (t_1 + t_2 + t_3)^r t_1(t_1 + t_2) - (t_2 + t_3)^r t_2(t_1) + t_3^r(t_1 + t_2)t_2$

$$= (t_1 + t_2 + t_3)^r t_1^2 + [((t_1 + t_2 + t_3)^r - (t_2 + t_3)^r + t_3^r)t_1 t_2 + t_3^r t_2^2] \geq 0$$

Example 6: If x, y, z are non-negative real numbers, prove that

(i) $x^2 + y^2 + z^2 \geq xy + yz + zx$

(ii) $x^3 + y^3 + z^3 + 3xyz \geq \sum_{sym} x^2 y = \sum_{cyc} (x^2 y + xy^2)$

(iii) If $abc = 1$, then $(a-1)(\frac{1}{b}-1) + (b-1)(\frac{1}{c}-1) + (c-1)(\frac{1}{a}-1) \geq 0$

(iv) $x^4 + y^4 + z^4 + xyz(x+y+z) \geq x^3 y + y^3 z + z^3 x + xy^3 + yz^3 + zx^3$

Note: Take any function $f(x, y, z)$ in three variables x, y and z .

1. $\sum_{cyc} f(x, y, z)$ means $f(x, y, z) + f(y, z, x) + f(z, x, y)$,

or, $\sum_{cyc} f(x_1, x_2, x_3)$ means $f(x_1, x_2, x_3) + f(x_2, x_3, x_1) + f(x_3, x_1, x_2)$.

2. $\sum_{sym} f(x, y, z)$ means $f(x, y, z) + f(y, z, x) + f(z, x, y) + f(x, z, y) + f(y, x, z) + f(z, y, x)$

or, $\sum_{cyc} f(x_1, x_2, x_3)$ means $f(x_1, x_2, x_3) + f(x_2, x_3, x_1) + f(x_3, x_1, x_2) + f(x_1, x_3, x_2) + f(x_2, x_1, x_3) + f(x_3, x_2, x_1)$.

3. $\sum_{cyc} x^3 y^2 = x^3 y^2 + y^3 z^2 + z^3 x^2$, $\sum_{sym} x^3 y^2 = x^3 y^2 + y^3 z^2 + z^3 x^2 + x^2 y^3 + y^2 z^3 + z^2 x^3$

Hence $\sum_{sym} x^3 y^2 = \sum_{cyc} (x^3 y^2 + x^2 y^3)$.

4. $\sum_{cyc} x^4 y^4 = x^4 y^4 + y^4 z^4 + z^4 x^4$, $\sum_{sym} x^4 y^4 = x^4 y^4 + y^4 z^4 + z^4 x^4 + x^4 y^4 + y^4 z^4 + z^4 x^4$

Hence $\sum_{sym} x^4 y^4 = 2 \sum_{cyc} x^4 y^4$.

Extension: If w, x, y, z and r are non-negative real numbers then $\sum_{cyc} w^r (w-x)(w-y)(w-z) \geq 0$.

- Example 7: If a, b, c are the lengths of the sides of a triangle, prove that

$$a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc. \quad (7^{\text{th}} \text{ IMO } 1964 \text{ No. } 2)$$
- Example 8: If x, y, z are positive real numbers such that $xyz = x + y + z + 2$, prove that

$$xy + yz + zx \geq 2(x + y + z).$$
- Example 9: If x, y, z are non-negative real numbers, prove that

$$3xyz + x^3 + y^3 + z^3 \geq 2[(xy)^{3/2} + (yz)^{3/2} + (zx)^{3/2}]$$
- Example 10: If a, b and c are non-negative real numbers, prove that

$$9(a^3 + b^3 + c^3) + 27abc \geq 2(a + b + c)^3$$
- Example 11: Let a, b, c be positive real numbers, prove that $(1 + \frac{4a}{b+c})(1 + \frac{4b}{c+a})(1 + \frac{4c}{a+b}) > 25$.
- Example 12: Let a, b, c be non-negative real numbers satisfying $a+b+c = 1$. Prove that

$$a^3 + b^3 + c^3 + 6abc \geq \frac{1}{4}.$$
 When does the equality hold?
- Example 13: Let a, b, c be non-negative real numbers satisfying $abc = 1$. Prove that

$$(a - 1 + \frac{1}{b})(b - 1 + \frac{1}{c})(c - 1 + \frac{1}{a}) \leq 1.$$
- Example 14: Let a, b, c be positive real numbers and p a real number. If $f(p) = abc \sum_{cyc} a^p$ and $g(p) = \sum_{cyc} a^{p+2}(b + c - a)$, determine which is larger, $f(p)$ or $g(p)$?
- Example 15: Given x, y, z are positive real numbers, prove that $\sum_{cyc} \frac{x^3 + xyz}{y+z} \geq x^2 + y^2 + z^2$.

Muirhead Inequality (米爾黑德不等式)

(Robert Franklin Muirhead, Scottish, 1860-1941)

Definition of Majorization (蓋 / 優化):

Given 2 n -tuples (元組), $(p_1, p_2, \dots, p_n)$ and $(q_1, q_2, \dots, q_n)$. If $p_1 \geq p_2 \geq p_3 \dots \geq p_n$, $q_1 \geq q_2 \geq q_3 \dots \geq q_n$, and they also satisfy the conditions $p_1 \geq q_1$, $p_1 + p_2 \geq q_1 + q_2$, ..., $p_1 + p_2 + \dots + p_{n-1} \geq q_1 + q_2 + \dots + q_{n-1}$ and $p_1 + p_2 + \dots + p_n = q_1 + q_2 + \dots + q_n$, then we say $(p_1, p_2, \dots, p_n)$ majorizes $(q_1, q_2, \dots, q_n)$ and we write $(p_1, p_2, \dots, p_n) \succ (q_1, q_2, \dots, q_n)$.

For example: (a). $(3, 0, 0) \succ (2, 1, 0)$,

$$(b). (p_1, p_2, \dots, p_n) \succ (\frac{p_1 + p_2 + \dots + p_n}{n}, \frac{p_1 + p_2 + \dots + p_n}{n}, \dots, \frac{p_1 + p_2 + \dots + p_n}{n})$$

on condition that $p_1 \geq p_2 \geq p_3 \dots \geq p_n$.

$$(c). (1, 0, 0, \dots, 0) \succ (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}) \quad \text{if both of them are } n\text{-tuples.}$$

$$(d). \text{For any triangle } ABC \text{ such that } \angle A \geq \angle B \geq \angle C, (\pi, 0, 0) \succ (A, B, C) \succ (\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3})$$

Definition of P-mean

Let $x_1, x_2, \dots, x_n$ be positive real numbers and $p = (p_1, p_2, \dots, p_n) \in \mathbb{R}^n$

The p -mean of $x_1, x_2, \dots, x_n$ is defined by $[p] = \left(\frac{1}{n!} \sum_{\sigma \in S_n} x_{\sigma(1)}^{p_1} x_{\sigma(2)}^{p_2} \dots x_{\sigma(n)}^{p_n} \right)^{\frac{1}{n}}$

Where S_n is the set of all permutations of $\{1, 2, \dots, n\}$, and there are $n!$ terms in the summation.

For example, let $p = (p_1, p_2, p_3)$ then $[p] = \frac{1}{3!} \sum_{sym} x^{p_1} y^{p_2} z^{p_3}$

$$= \frac{1}{3!} (x^{p_1} y^{p_2} z^{p_3} + y^{p_1} z^{p_2} x^{p_3} + z^{p_1} x^{p_2} y^{p_3} + x^{p_1} z^{p_2} y^{p_3} + y^{p_1} x^{p_2} z^{p_3} + z^{p_1} y^{p_2} x^{p_3})$$

More examples: (a) Let $p = (2, 1)$, then $[p] = \frac{1}{2}(x^2 y + y^2 x)$

$$(b) \text{ Let } p = (1, 3, 2), \text{ then } [p] = \frac{1}{3!}(xy^3 z^2 + yz^3 x^2 + zx^3 y^2 + xz^3 y^2 + yx^3 z^2 + zy^3 x^2)$$

$$(c) \text{ Let } p = (1, 3, 1), \text{ then } [p] = \frac{1}{3!}(xy^3 z + yz^3 x + zx^3 y + xz^3 y + yx^3 z + zy^3 x)$$

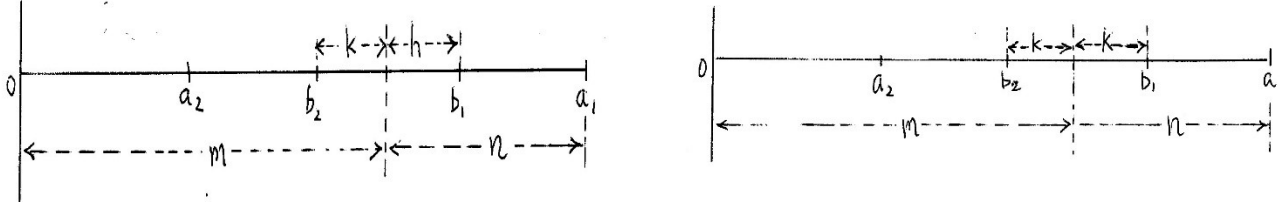
$$(d) \text{ Let } p = (0, 3, 1), \text{ then } [p] = \frac{1}{3!}(y^3 z + z^3 x + x^3 y + z^3 y + x^3 z + y^3 x)$$

Theorem: Let $x_1, x_2, \dots, x_n$ be positive real numbers and $p, q \in \mathbb{R}^n$. If $p \succcurlyeq q$, then $[p] \geq [q]$.

Furthermore, for $p \neq q$, equality holds if and only if $x_1 = x_2 = \dots = x_n$.

Proof for n=2: For any positive real numbers x, y and $(a_1, a_2), (b_1, b_2) \in \mathbb{R}^2$ such that $(a_1, a_2) \succcurlyeq (b_1, b_2)$, we want to prove that $\sum_{sym} x^{a_1} y^{a_2} \geq \sum_{sym} x^{b_1} y^{b_2}$.

Proof: Given $(a_1, a_2) \succcurlyeq (b_1, b_2)$, which means that $a_1 \geq a_2, b_1 \geq b_2$, and $a_1 \geq b_1, a_1 + a_2 = b_1 + b_2$.



$$\text{Let } m = \frac{a_1 + a_2}{2}, n = \frac{a_1 - a_2}{2}, h = (b_1 - m), k = (m - b_2).$$

Then we have $a_1 = m + n, a_2 = m - n, b_1 = m + h, b_2 = m - k$

Further, since $a_1 + a_2 = b_1 + b_2$, hence $(m + n) + (m - n) = (m + h) + (m - k)$

So we have $h = k$. Finally $b_1 = m + k, b_2 = m - k$

$$\begin{aligned} \text{Then } \sum_{sym} x^{a_1} y^{a_2} - \sum_{sym} x^{b_1} y^{b_2} &= \sum_{sym} (x^{a_1} y^{a_2} - x^{b_1} y^{b_2}) \\ &= x^{a_1} y^{a_2} + y^{a_1} x^{a_2} - x^{b_1} y^{b_2} - y^{b_1} x^{b_2} \\ &= x^{m+n} y^{m-n} + y^{m+n} x^{m-n} - x^{m+k} y^{m-k} - y^{m+k} x^{m-k} \\ &= x^{m-n} y^{m-n} (x^{2n} + y^{2n} - x^{n+k} y^{n-k} - x^{n-k} y^{n+k}) \\ &= x^{m-n} y^{m-n} (x^{n+k} - y^{n+k})(x^{n-k} - y^{n-k}) \geq 0 \\ &\quad (\text{since } (n+k) \text{ and } (n-k) \text{ are both positive}). \end{aligned}$$

$$\text{Hence } \sum_{sym} x^{a_1} y^{a_2} \geq \sum_{sym} x^{b_1} y^{b_2}, \text{ and}$$

$$[(a_1, a_2)] = \frac{1}{2!} \sum_{sym} x^{a_1} y^{a_2} \geq \frac{1}{2!} \sum_{sym} x^{b_1} y^{b_2} = [(b_1, b_2)].$$

Proof for n=3: For any real numbers x, y, z and $(a, b, c), (p, q, r) \in \mathbb{R}^3$ such that $(a, b, c) \succcurlyeq (p, q, r)$, we want to prove that $\sum_{sym} x^a y^b z^c \geq \sum_{sym} x^p y^q z^r$.

Proof: Here we have $a \geq b \geq c, p \geq q \geq r, a \geq p, a + b \geq p + q$, and $a + b + c = p + q + r$.

$$\begin{aligned} \text{Case 1: } p \geq b, \quad \sum_{sym} x^a y^b z^c &= \sum_{cyc} z^c (x^a y^b + x^b y^a) \\ &\geq \sum_{cyc} z^c (x^p y^{a+b-p} + x^{a+b-p} y^p) \\ &= \sum_{sym} x^p y^{a+b-p} z^c \\ &= \sum_{cyc} x^p (y^{a+b-p} z^c + y^c z^{a+b-p}) \\ &\geq \sum_{cyc} x^p (y^q z^r + y^r z^q) \\ &= \sum_{sym} x^p y^q z^r \end{aligned}$$

$$\begin{aligned} \text{Case 2: } p < b, \quad \sum_{sym} x^a y^b z^c &= \sum_{cyc} x^a (y^b z^c + y^c z^b) \\ &\geq \sum_{cyc} x^a (y^p z^{b+c-p} + y^{b+c-p} z^p) \\ &= \sum_{sym} x^a y^p z^{b+c-p} \\ &= \sum_{cyc} y^p (x^a z^{b+c-p} + x^{b+c-p} z^a) \\ &\geq \sum_{cyc} y^p (x^q z^r + x^r z^q) \\ &= \sum_{sym} x^p y^q z^r \end{aligned}$$

In both cases, $\sum_{sym} x^a y^b z^c \geq \sum_{sym} x^p y^q z^r$.

More illustrations: (a): $(2,0) \prec (1,1) \Rightarrow \frac{1}{2!} \sum_{sym} x^2 y^0 \geq \frac{1}{2!} \sum_{sym} x^1 y^1 \Rightarrow \frac{x^2 + y^2}{2} \geq xy$

(b): $(3,0,0) \prec (1,1,1) \Rightarrow \frac{1}{3!} \sum_{sym} x^3 y^0 z^0 \geq \frac{1}{3!} \sum_{sym} xyz \Rightarrow \frac{x^3 + y^3 + z^3}{3} \geq xyz$

Example 16: Consider the case of three given positive real numbers x, y and z , prove that
 (i) R.M.S. $\geq$ A.M. (ii) G.M. $\geq$ H.M.

Example 17: Prove that $2(x^5 + y^5) \geq (x^2 + y^2)(x^3 + y^3)$.

Example 18: For any positive real numbers a, b and c , prove that $(a+b)(b+c)(c+a) \geq 8abc$.

Example 19: For any positive real numbers a, b and c , prove that $\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c$.

Example 20: For any positive real numbers a, b and c , prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$.

Example 21: For any positive real numbers a, b and c , prove that $\frac{a+b}{c^2} + \frac{b+c}{a^2} + \frac{c+a}{b^2} \geq 2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

Example 22: For any real a, b and c , such that $a^2 + b^2 + c^2 = 1$, prove $3(a^2bc + ab^2c + abc^2) \leq 1$

Example 23: For any positive real numbers satisfying $abc = 1$, prove that

$$\frac{1}{a+b+1} + \frac{1}{b+c+1} + \frac{1}{c+a+1} \leq 1.$$

Example 24: For any positive real numbers a, b and c , prove that

$$\frac{a^3+b^3}{(a+b)^3} + \frac{b^3+c^3}{(b+c)^3} + \frac{c^3+a^3}{(c+a)^3} + \left(\frac{27}{4}\right)\left(\frac{abc}{a^3+b^3+c^3}\right) \leq 3.$$

Example 25: For any non-negative real numbers x, y, z satisfying $x+y+z=1$, prove that

$$0 \leq xy + yz + zx - 2xyz \leq \frac{7}{27} \quad (25^{\text{th}} \text{ IMO, } 1984 \text{ No. } 1)$$

Note: If the product $x_1, x_2, \dots, x_n$ is 1, then $[(p_1, p_2, \dots, p_n)] = [(p_{1-r}, p_{2-r}, \dots, p_{n-r})]$

Proof: $[(p_{1-r}, p_{2-r}, \dots, p_{n-r})] = \left(\frac{1}{n!}\right) \sum_{\sigma \in S_n} x_{\sigma(1)}^{p_{1-r}} x_{\sigma(2)}^{p_{2-r}} \dots x_{\sigma(n)}^{p_{n-r}}$

$$= \left(\frac{1}{n!}\right) \sum_{\sigma \in S_n} \frac{x_{\sigma(1)}^{p_1} x_{\sigma(2)}^{p_2} \dots x_{\sigma(n)}^{p_n}}{(x_{\sigma(1)} x_{\sigma(2)} \dots x_{\sigma(n)})^r}$$

$$= \left(\frac{1}{n!}\right) \sum_{\sigma \in S_n} x_{\sigma(1)}^{p_1} x_{\sigma(2)}^{p_2} \dots x_{\sigma(n)}^{p_n}$$

$$= [(p_1, p_2, \dots, p_n)]$$

Example 26: Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}. \quad (36^{\text{th}} \text{ IMO, } 1995 \text{ No. } 2)$$

Example 27: For any $x, y, z > 0$ with $xyz \geq 1$, prove that

$$\frac{x^5-x^2}{x^5+y^2+z^2} + \frac{y^5-y^2}{x^2+y^5+z^2} + \frac{z^5-z^2}{x^2+y^2+z^5} \geq 0 \quad (46^{\text{th}} \text{ IMO (2005) No. } 3)$$

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